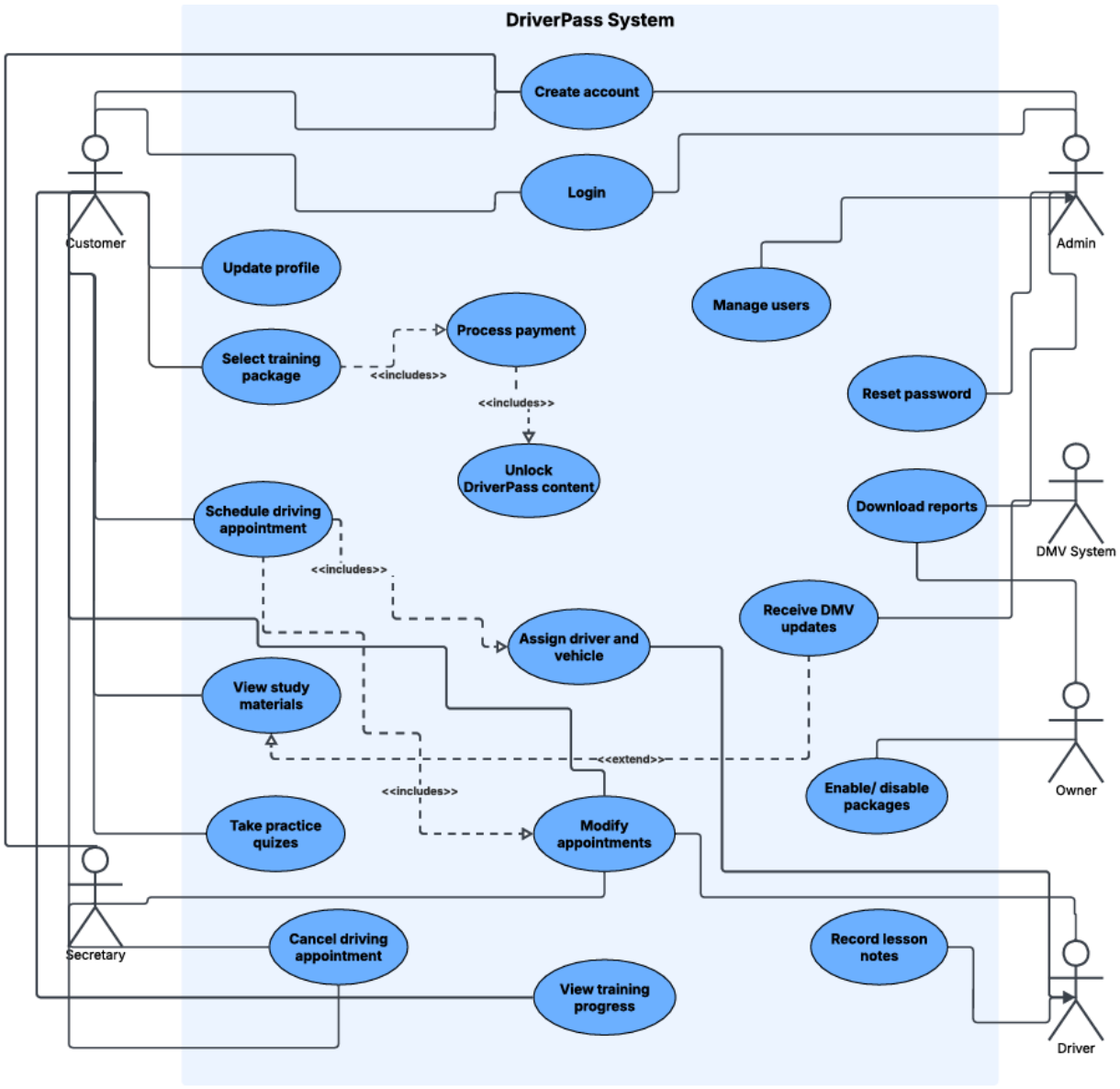
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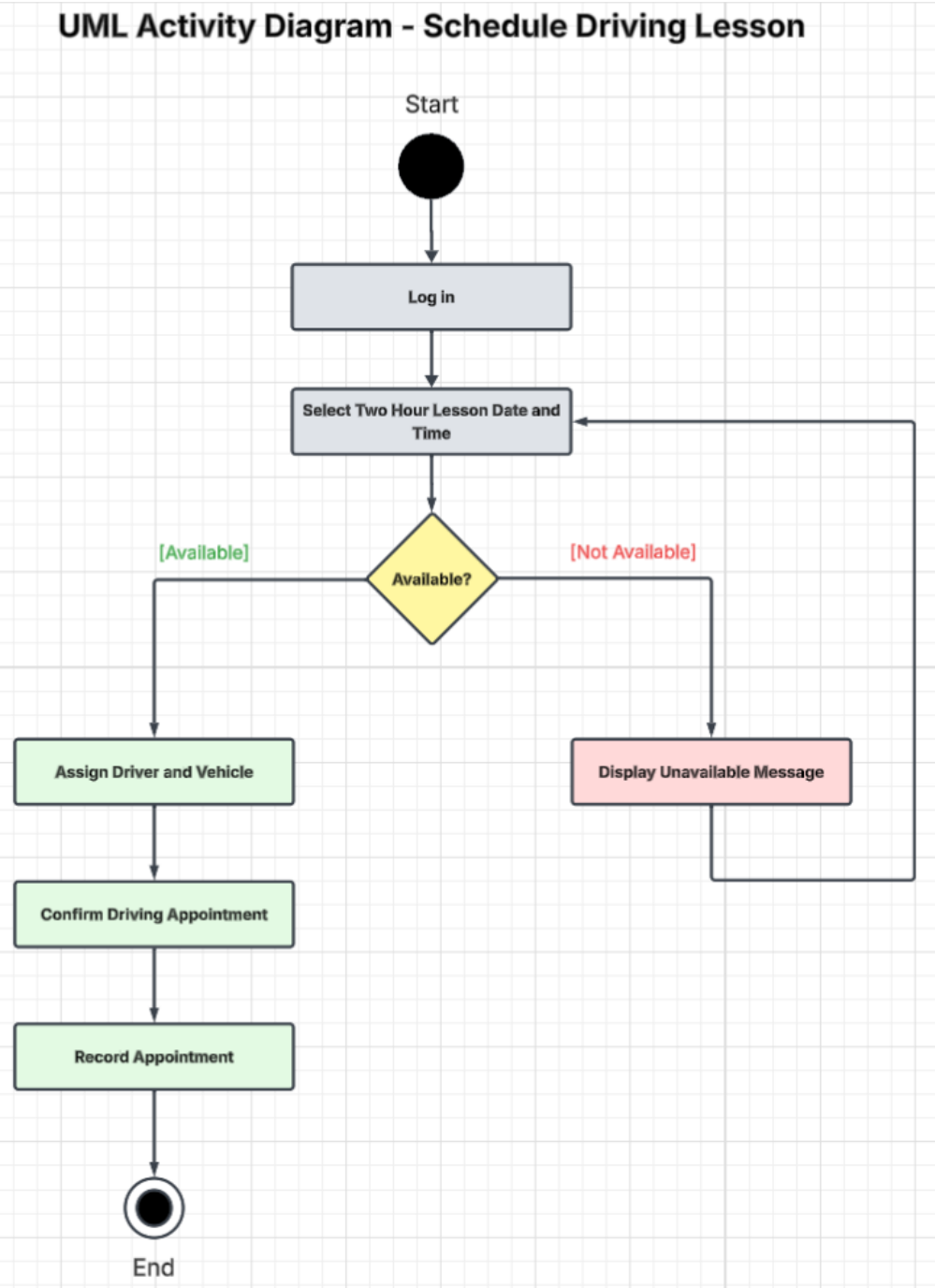
CS 255 System Design Document

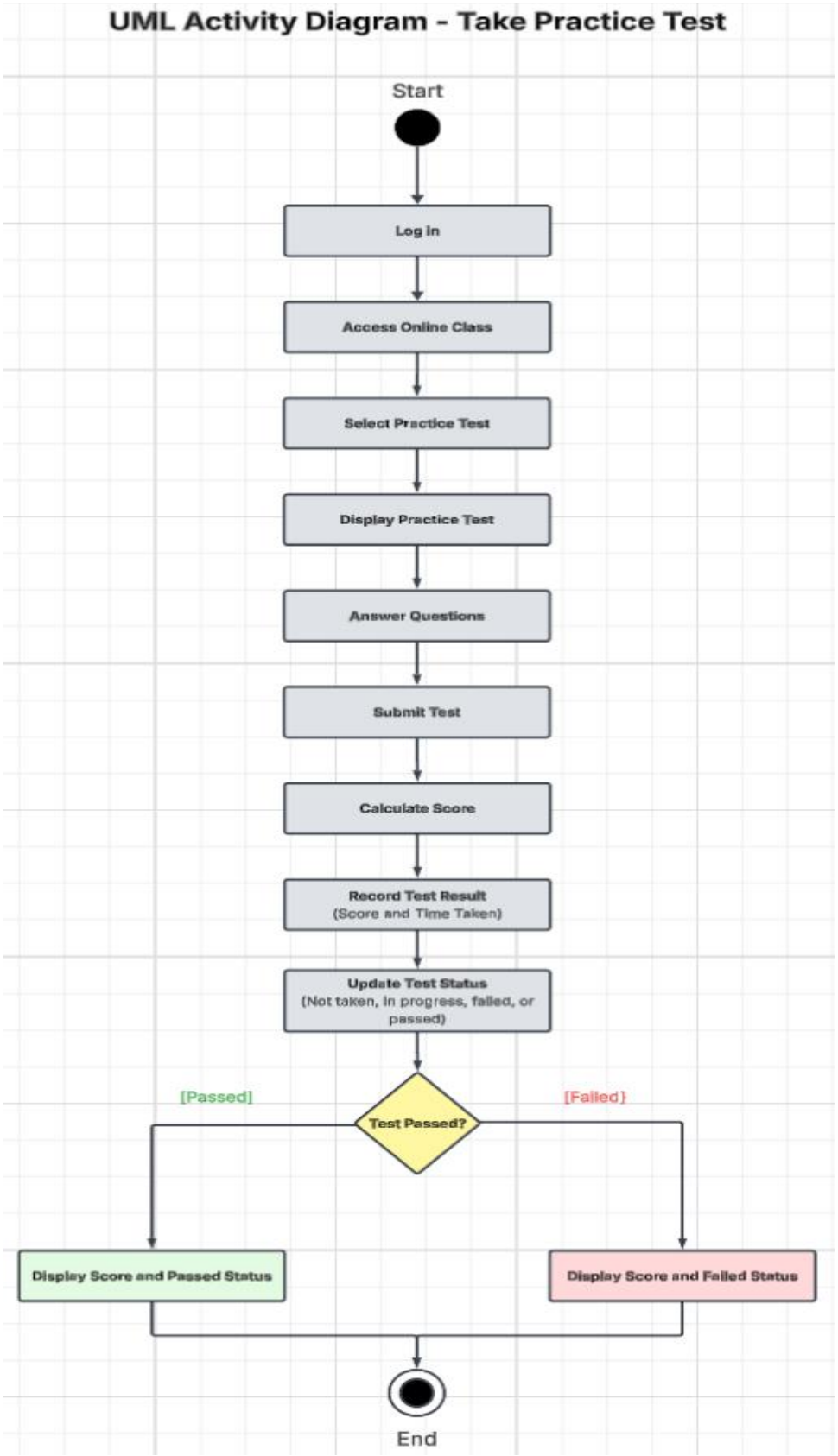
UML Diagrams

UML Use Case Diagram

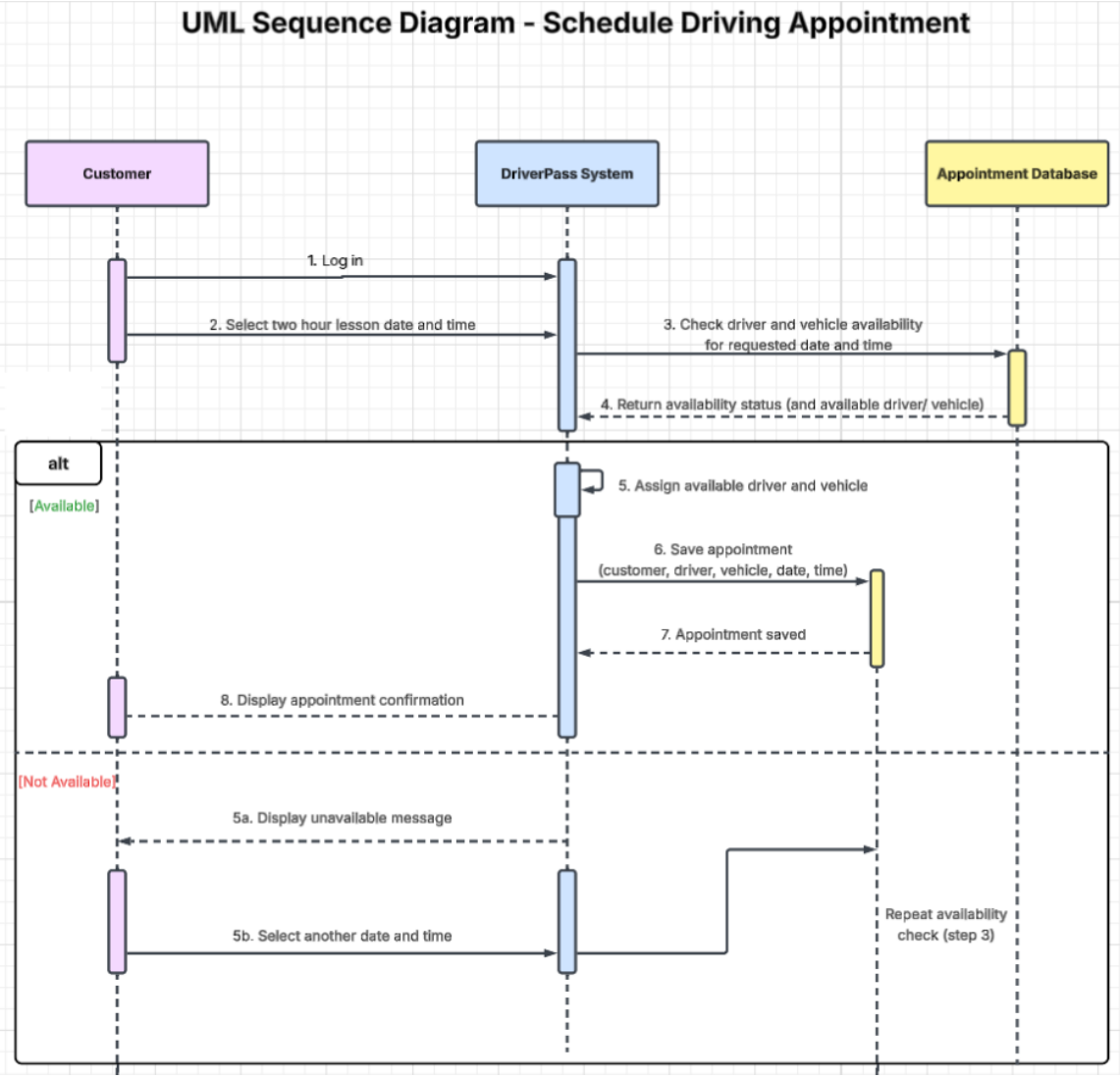


UML Activity Diagrams

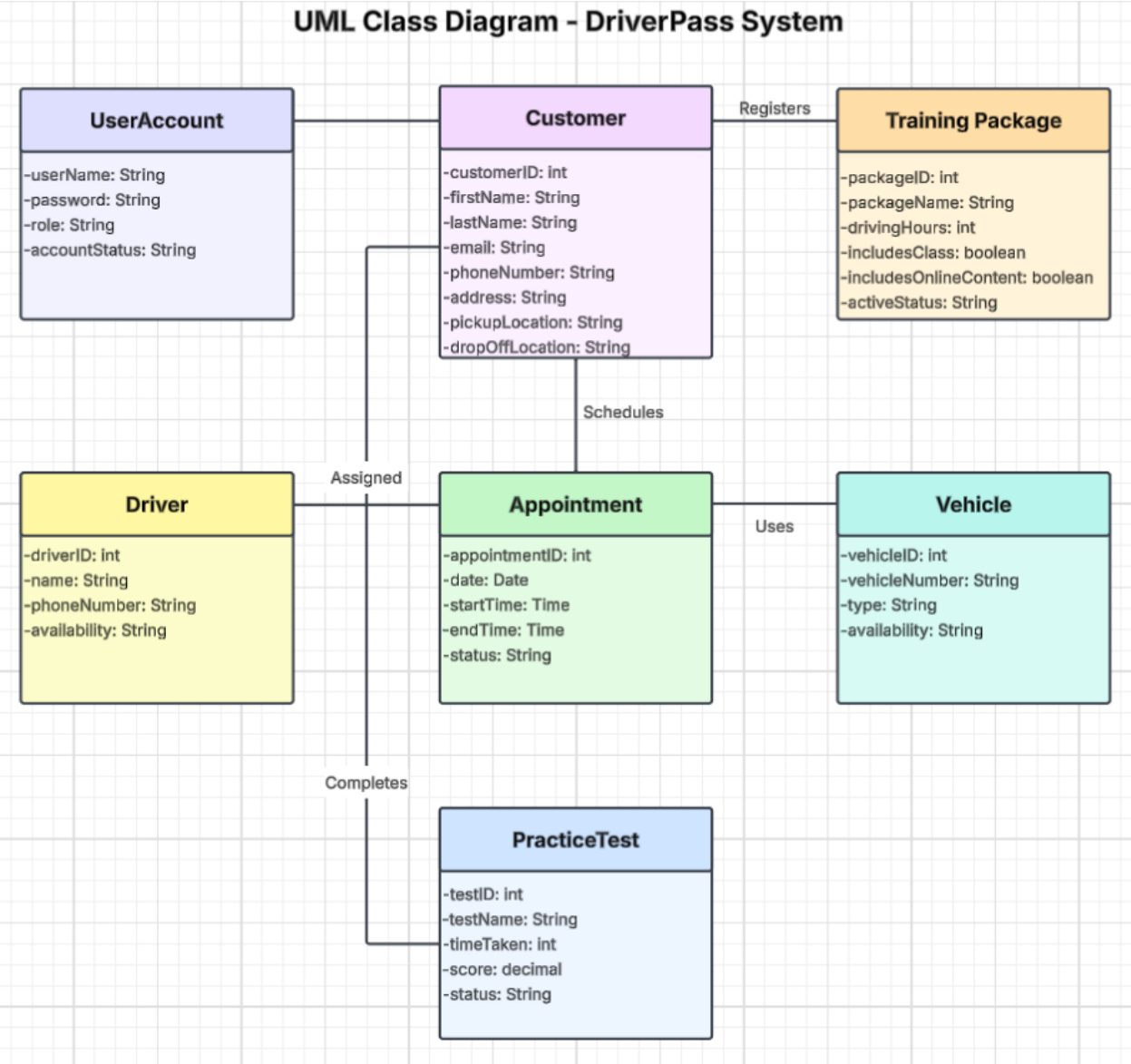



UML Activity Diagram - Take Practice Test
Start
Log in
Access Online Class
Select Practice Test
Display Practice Test
Answer Questions
Submit Test
Calculate Score
Record Test Result
(Score and Time Taken)
Update Test Status
(Not taken, In progress, failed, or passed)
[Passed]
Test Passed?
[Failed]
Display Score and Passed Status
Display Score and Failed Status
End

UML Sequence Diagram



UML Class Diagram



Technical Requirements

- **Hardware:** DriverPass employees will need computers that have stable internet connections to handle customer accounts, appointments, training records, and reports. Customers should be able to reach the system from both computers and mobile devices using an internet connection.

- **Software:** The platform will need a cloud-based web application that users can open in a web browser. It should include secure login, user access based on roles, password management, customer account handling, training package selection, appointment booking, changes and cancelations, online training, practice exams, progress monitoring, and activity reporting.

- **Database and Tools:** There must be a central database to keep customer and staff accounts, training packages, appointments, driver and vehicle assignments, test outcomes, and other system records. Reporting tools should let approved staff create and download activity reports in a format that works with programs like Microsoft Excel.

- **Security:** All connections between users and the DriverPass system should be secured to keep information safe as it moves between the client and server. Access should depend on user roles, and the system must protect accounts from repeated failed login attempts while offering a secure password reset process.

- **Infrastructure:** DriverPass will need cloud hosting to allow centralized access, data storage, backups, security, and system uptime. The system should also be able to communicate with the DMV so DriverPass can get updates and keep current rules, policies, and practice test details.